

Interaction, Improvisation, and Interplay in Jazz

Robert Hodson



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Hope College

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Preface

This book arises from the confluence of my experiences as a performing musician and a music scholar. Originally a classical pianist, I also developed a strong interest in jazz, cultivated through listening to recordings, radio broadcasts, and attending live performances. I was particularly fascinated with and mystified by both the improvisational abilities of jazz musicians as well as the way that the members of small jazz ensembles coordinated their individual parts into cohesive, coherent performances.

As my increasing interest in jazz led me to a hands-on study of jazz improvisation and performance, I had the great fortune to encounter some excellent teachers as well as the priceless opportunity to perform weekly with some really fine, professional players. As I became more and more comfortable playing with these musicians, I began to notice interesting things going on around me during these performances. That is, I began to hear that the musicians were not just *coordinating* their individual improvised parts, but that there was an amazing amount of interaction going on between the players; the musicians were frequently responding to one another, engaging in musical conversations in which each musician's improvisation was, at times, affected and influenced by the other musicians' improvisations. At first, I was able to do little more than notice these interactions taking

place around me, but as I gained experience, I found I was able to participate in these interactions. Through this experience, I gained a real appreciation for the dynamic, interactive processes that take place between jazz musicians, and realized that these interactions are an essential part of a jazz performance.

While pursuing a doctorate in music theory at the University of Wisconsin–Madison, I decided that I wanted to focus my research on jazz, and I began to survey the theoretical and analytical writings on jazz improvisation. As I read article after article, I noticed that the authors tended to focus on single improvised lines, rarely considering the possibility that the simultaneously improvised parts of the other members of the ensemble could have an effect on the solo line they were analyzing. In other words, they tended to ignore the dynamic, interactive processes that I found—and continue to find—to be so interesting in jazz performances. This book is my attempt to expand the scope of jazz theory and analysis beyond this narrow focus on a single improvised line. To this end, the first chapter develops a model of the improvisational process that includes player interaction as a fundamental principle, and all of my analyses will be built around the idea that jazz musicians do not improvise in isolation, and that a jazz performance is as much about what happens *between* musicians as it is about each musician's individual improvisation.

This book began as my doctoral dissertation, and I would particularly like to thank my adviser, Brian Hyer, for his guidance throughout that project. His tremendous generosity with his time and his careful reading and thoughtful comments on my many drafts had a profound effect on the final form of the dissertation and, as a result, this book. Brian provided me with an exemplary model of what it means to be a scholar, teacher, and mentor, and I hope to do for my students what he did for me.

I am currently on the faculty of Hope College in Holland, Michigan, and I would like to thank the college for supporting my work on this project. Provost James Boelkins and Dean for Arts and Humanities William Reynolds deserve special thanks for supporting this project through summer research grants and through assisting with publication expenses. Hope College is a great place to teach and work and I would like to thank my colleagues and my students in the

music department for their questions, ideas, support, feedback, and positive energy.

I would like to acknowledge and express gratitude to those who taught me about jazz and how to play it. Lazaro Vega of WBLV Blue Lake Public Radio was my first jazz teacher; his nightly radio shows introduced me to jazz and taught me so much about this incredible music. Bassist Richard Davis also deserves special mention; his mentoring and friendship over the last dozen or so years has influenced me greatly. I'm also grateful to Ron Newman, Peter Dominguez, Marcus Belgrave, Eddie Russ, Andrew Speight, and Mtafiti Imara for their roles in my jazz education and performing experience.

Finally, I would like to thank my wife, Carrie, for her boundless love and support. To her, and to our children, Isabel and Simon, I dedicate this book with love and affection.

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JAZZ IMPROVISATION

Theory, Analysis, Context, and Process

Most technical writings on jazz focus on improvised lines and their underlying harmonic progressions. These writings often overlook the basic fact that when one listens to jazz, one almost never hears a single improvised line, but rather a texture, a musical fabric woven by several musicians in real time. While it is often pragmatic to single out an individual solo line (I will be doing the same), it is important at all times to remember that an improvised solo is but one thread in that fabric, and it is a thread supported by, responded to, and responsive of the parts being played by the other musicians in the group. This book will explore the process of player interaction in jazz, and the role this interaction plays in generating improvised music.

I'd like to begin by examining a recorded excerpt from an improvisation by saxophonist Cannonball Adderley. To some extent, recordings in jazz take the place—and assume the same textual authority—of scores in European music, and I will rely on them throughout this work. Furthermore, I will make extensive use of transcriptions in my analyses, and while the authority of transcription has been widely critiqued in ethnomusicological circles,¹ I intend to circumvent that issue by viewing transcriptions as heuristics in my discussions of recorded performances. In other words, my transcriptions are meant to serve as visual aids to the recordings and to my commentaries on them. I do not assume that they capture all aspects of a recorded performance, and I will be routinely taking information into account that is not available in the transcription. I regard them as representations of performances rather than pieces, and at times those representations will be approximate—some of the music I will be dealing with is extremely complicated by the standards of any music notational

tradition. It is important to remember that in jazz, a score (whether a “lead sheet,” arrangement, or transcription) occupies no privileged ontological position, a notion that I take as an article of faith; my use of them is entirely pragmatic.

The excerpt, transcribed in figure 1.1, comes from the end of Adderley’s first improvised chorus on the jazz standard “Groovin’ High.”² As a starting point, the analysis will first examine Adderley’s improvised line by itself, before demonstrating that an examination of the musical context within which the improvisation took place can generate further insight. The analysis will introduce and briefly address a number of musical and analytical issues to be dealt with at length in later chapters.

Rests in mm. 27 and 28 suggest a preliminary parsing of the improvisation into three large gestures, which are bracketed in figure 1.1 and labeled A, B, and C. Each gesture is of a different length: A is nine beats long, beginning in m. 25 and extending through the downbeat of m. 27; B is approximately four beats long, beginning with a pick-up to beat 3 of m. 27 and extending to m. 28, beat 2; and C, the longest of the three, extends approximately fourteen beats from a pick-up to beat 4 of m. 28 through the downbeat of m. 32.

At first glance, comparing the scalar sixteenth-note improvised lines of gestures A and B with the somewhat less-scalar eighth-note line of gesture C suggests that this excerpt is made up of two quite different kinds of music. The following analysis supports that preliminary intuition, first describing the differences between gestures A–B and gesture C, and then suggesting possible reasons for these differences.

Besides sharing a preponderance of sixteenth notes, gestures A and B have many other similarities. As noted earlier, they are primarily scalar—that is, they are almost exclusively composed of stepwise motion—and both gestures are highly chromatic. A and B are also similar in another, less obvious, way in that both the contour and the melodic pitches of the improvised lines relate to the underlying harmony. Comparing gestures A and B with the underlying harmony reveals some consistent harmonic processes. Figure 1.2 presents a “harmonic analysis” of these melodic gestures. The upper staff adds analytical beams to the transcription, the middle staff gives a crude, “close position” realization of the chord symbols; these chords are

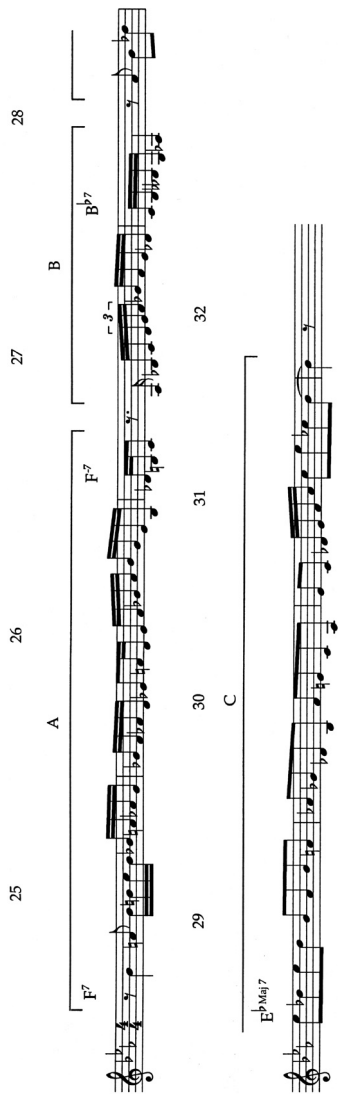


Figure 1.1. Cannonball Adderley, “Groovin’ High,” saxophone solo, mm. 25–32.

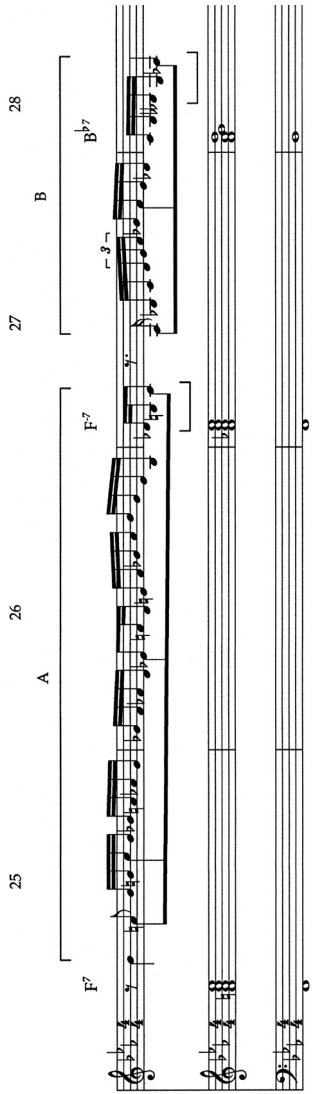


Figure 1.2. Cannonball Adderley, “Groovin’ High,” saxophone solo, mm. 25–28.

presented over a simple realization of the chord progression's root movement on the lower staff. The chord symbols in this figure are standard in jazz performance practice, and are interpreted as follows: The superscript ⁻⁷ denotes a minor seventh chord, constructed from a minor triad and minor 7th; the symbol C⁻⁷ would therefore designate the chord containing the pitches C, E $\flat$, G, and B $\flat$. The plain superscript ⁷ denotes a "dominant" seventh chord, made up of a major triad and minor 7th; C⁷ would therefore be C–E–G–B $\flat$. The superscript ^{Maj7} designates a major seventh chord, containing a major triad and major 7th; C^{Maj7} would thus be spelled C–E–G–B. Finally, these chords may contain additional alterations: C^{-7(b5)}, for instance, would designate a C minor seventh chord with a flatted-5th, or C–E $\flat$ –G $\flat$ –B $\flat$.

As the added beams in figure 1.2 demonstrate, the contour of the improvised line emphasizes harmonically significant pitches: after the initial B $\flat$ moves down to A (the 3rd of F⁷), the line ascends to its high point, C (the 5th), before descending chromatically to E $\flat$ (the 7th). E $\flat$ is even more strongly emphasized than the previous harmonic pitches A and C for several reasons: (1) as an eighth note, E $\flat$ receives an agogic accent after the long string of sixteenths; (2) it also receives a metric accent as a result of its falling *on* the beat; and (3) it is approached by double chromatic neighbors—F $\flat$ above and D $\sharp$ below. As the two brackets below the upper staff in figure 1.2 point out, Adderley employs this double chromatic neighbor figure motivically, transposing it just after the downbeat of m. 27 to embellish C (the 5th of F⁷) at the end of the first gesture and again in m. 28 to embellish A $\flat$ (the 7th of B $\flat$ ⁷) at the end of gesture B. These double neighbors, when combined with the following rests, bring about a sense of melodic closure, and help define the endings of these gestures. Gesture B not only ends with the same closing motive as gesture A, but also like A emphasizes harmonically significant pitches through its contour, beginning on A $\flat$ (the 3rd of F⁻⁷), ascending to an embellished F (the root of F⁷), passing through B $\flat$ (the root of B $\flat$ ⁷), before finally descending to A $\flat$ (the 7th of B $\flat$ ⁷).

Another similarity between gestures A and B worth noting is the rhythmic placement of these harmonically significant pitches. Without exception, all of these pitches occur in relatively weak rhythmic positions—either off the beat (as in the first two beamed pitches of

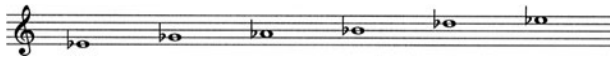


Figure 1.3. E $\flat$ minor pentatonic scale.



Figure 1.4. Cannonball Adderley, “Groovin’ High,” saxophone solo, mm. 29–32.

each gesture) or, if on the beat, on the metrically weak second beat of the measure (as in the third beamed pitch of each gesture). This rhythmic displacement of harmonic pitches keeps the music off balance and pushes it forward, making the improvised melody seem as if it floats over the meter, rather than being constrained by it.

Gesture C contrasts markedly with gestures A and B. Immediately noticeable is the change from sixteenth to eighth notes. Also noticeable is the fact that the melody of gesture C is not as consistently stepwise as the melodies in A and B, but contains a greater variety of melodic intervals. This contrast is, in part, a result of the melodic resources used in each gesture: A and B are constructed from a combination of chromatic and diatonic scales that Adderley uses to fill in gaps between chord tones, while C is constructed primarily from an E $\flat$ minor pentatonic scale, written out in figure 1.3. The minor pentatonic scale is frequently used by improvisers to give the music a “bluesy” or “funky” feel. This bluesy feel derives from the use of “blue” notes—primarily the flatted 3rd G $\flat$ and 7th D $\flat$, which in figure 1.4 clash colorfully with the G $\sharp$ and D $\sharp$ of the underlying E $\flat$ ^{Maj7}.

Figure 1.4 shows not only the clashes that result from playing the E $\flat$ minor pentatonic scale over E $\flat$ ^{Maj7}, but also that gesture C differs from A and B in the rhythmic placement of its harmonic pitches. Whereas in figure 1.2 we saw that chromatic neighbors displaced harmonic pitches to relatively weak rhythmic positions throughout A and B, harmonic pitches in C regularly occur *on* the beat, frequently right on the *first* beat of the measure. This has the effect of “rooting” the melody to the

	Gestures A-B	Gesture C
Rhythmic values:	faster (sixteenths)	slower (eighths)
Rhythmic subdivision:	more even (straight)	less even (swing)
Meter:	de-emphasized	emphasized
Melody:	more scalar	less scalar
Melodic resources:	chromatic scale, modes	minor pentatonic scale
Pitch:	more accurate/ less inflected	less accurate/ more inflected
Harmonic pitch placement:	more syncopated	less syncopated

Figure 1.5. Comparison of gestures A–B and gesture C.

beat, binding it to the meter, in contrast to the way the melody breaks free and hovers above the meter in gestures A and B.

In addition to the differences discussed so far, two more very subtle differences exist between gestures A–B and gesture C. These differences—which are difficult if not impossible to capture in standard musical notation—have to do with subtle manipulations of pitch and rhythm. In gesture C, pitches tend to be *inflected*—that is, they may go in and out of tune, or rather “slid up” to, “squeezed,” or “smeared,” to use common jazz lingo—in contrast to the precise, in-tune pitches of gestures A and B. Subdivisions of the beat in A and B are also more precise than that of C: in A and B, the melody divides the beat into four *very* even sixteenths, while in C, the beat is divided into two *uneven* notes, the first longer and the second shorter. In jazz slang, the sixteenths in gestures A and B are played “straight,” while Adderley “swings” the eighths in C. The long/short pattern of the swing eighths further emphasizes the beat in gesture C, increasing the difference in “feel” between it and gestures A and B.

Figure 1.5 summarizes some of the crucial differences between gestures A–B and gesture C. What intuition suggested *before* the analysis is confirmed by the analysis: the improvised line of gesture C is a different *kind* of music than that of gestures A and B—it is constructed differently and operates under different principles. The characteristics of these two different kinds of music correspond to two different styles of jazz: the first style, used in gestures A and B, is heard as *bebop*, and the second style, that used in gesture C, is heard as *blues*. So, between gestures B and C, Adderley shifts musical gears, switching from one style to another.

Up to this point, the analysis merely describes *how* Adderley switches from one style to another. A more interesting question is

“*Why* does Adderley make the switch from bebop to blues?” A number of explanations might be offered. Maybe after the long string of sixteenthths, Adderley sensed a need for variety. Maybe he wanted to increase the energy level at the end of the chorus, but had reached his upper limit with the sixteenth-note lines, and switched to the more “vocal” blues style to make a stronger rhetorical statement. Maybe there isn’t a strictly musical reason, and the change was just the result of a whim. Any of these reasons could be argued, but an additional possibility that can be even more strongly argued appears upon examination of the musical context within which the line was improvised.

Adderley’s improvisation takes place in the context of a small-group jazz performance. A small jazz group usually consists of a rhythm section (piano, bass, and drums) and one or more “front-line” instruments, usually saxophone, trumpet, or trombone. The ensemble on this recording features a rhythm section made up of Wynton Kelly, piano, Percy Heath, bass, and Art Blakey, drums, and these are the musicians who accompany Adderley’s improvisation. Even though standard jazz performance practice assigns certain musical roles to each member of the rhythm section (the drummer plays rhythmic patterns, the bassist “walks” a quarter-note line that defines the harmonic progression, and the pianist plays syncopated chords, or “comps”), the way that the members choose to fulfill these roles is flexible and often spontaneously negotiated in performance (rhythm section roles are discussed in more detail later in this chapter). In other words, *all of the members of the ensemble are improvising simultaneously*; the improvised saxophone melody is supported by the rhythm section’s equally improvised accompaniment.

Examining Wynton Kelly’s comping suggests another possible explanation for Adderley’s change in style between gestures B and C. Through A and B, Kelly’s comping is understated and fairly standard—he plays light, syncopated chords, providing rhythmic and harmonic support to Adderley’s solo. However, in the break between Adderley’s gestures B and C, he inserts a stronger, louder, fuller chord—in effect, he dramatically *punctuates* Adderley’s improvisation. This particular chord “voicing,” notated in figure 1.6, is remarkably colorful. Besides containing the root B♭, 3rd D, 7th A♭, and 13th G of B♭⁷, it also contains the blue notes D♭ and G♭ that clash with these chord members.

Figure 1.6 shows a musical score for measures 27-29 of "Groovin' High". The score is for Saxophone and Piano. Measure 27 has a B chord. Measure 28 has a B $\flat$ 7 chord. Measure 29 has a E $\flat$ Maj7 chord. A box highlights the piano accompaniment in measure 28, showing a voicing with D, G, and D $\flat$.

Figure 1.6. "Groovin' High," mm. 27–29.

The most common blue notes arise from the clash between major and minor 3rds. In this voicing, both D, the major 3rd of the B $\flat$ 7 chord, and G, the third scale-degree in E $\flat$ major, are played at the same time as their blue notes D $\flat$ and G $\flat$. With this dramatically placed, very bluesy chord voicing, Kelly asserts or "signifies" the blues.³

Kelly's strong, bluesy voicing of this chord seems to have a galvanizing effect on the performance. It's almost as if Adderley hears this chord, draws upon the knowledge of jazz styles he shares with Wynton Kelly, realizes that this harmonic configuration signifies the blues style, and responds to Kelly's assertion by continuing his improvisation in a blues idiom. This type of interaction might be compared to a conversation: Kelly suggests a "topic" (the blues style), Adderley "responds" by elaborating on the topic. For the conversational metaphor to hold, though, the communication between the players would of course have to flow in both directions; further reflection on this performance shows this to be the case.

Figure 1.7 continues the transcription into the beginning of the next chorus. Adderley begins this chorus by introducing a two-note motive—the tonic E $\flat$ preceded by its leading tone D—which he repeats and

Figure 1.7 shows a musical score for measures 33-35 of "Groovin' High". The score is for Saxophone and Piano. Measure 33 has an E $\flat$ 7 chord. Measure 34 has an A-7 chord. Measure 35 has an A-7 chord. A box highlights the piano accompaniment in measure 33, showing a voicing with D and E $\flat$.

Figure 1.7. "Groovin' High," mm. 33–35.

develops over the next few measures. Kelly hears this simple, yet strong, motivic idea and responds immediately, adopting the D–E $\flat$ neighbor figure as the uppermost notes in his chord voicing. Over the next few measures, both Adderley's and Kelly's improvisations are controlled by this motive.

This analysis shows that communication and interaction between players in small-group jazz can and frequently does affect an individual musician's improvisation. Indeed, the ability to listen and respond to one's bandmates is highly valued among jazz musicians. This dynamic interaction between simultaneously improvising musicians is the subject of this book: while most analyses of jazz improvisation tend to focus on the structure of an individual musician's improvised line, my analyses will consider not only these individual improvisations, but also the structure of interaction *between* the musicians in an ensemble, and will explore the ways this interaction can influence each individual musician's improvisation.

On the Nature and Analysis of Jazz Improvisation and Performance

The analysis of jazz improvisation presents some difficulties. First, since most of a jazz performance is improvised, there is no score to examine; typically, an analyst will want to create a transcription to aid the discussion of a recorded performance. Besides this practical difficulty, analyzing improvisation raises a number of larger issues, alluded to above, including the following: Is a transcribed improvisation comparable to a composed score? Are existing analytical models, which were devised for the analysis of composed scores, appropriate for the examination of an improvised solo? What should be our goal in analyzing improvised music? In other words, what is the nature of jazz improvisation and performance?

With regard to the issue of whether a transcribed improvisation is comparable to a composed score and can be analyzed as such, a number of authors express different viewpoints. These viewpoints can be boiled down to the question of whether an improvisation is viewed as a product or a process. If the improvisation is viewed as a product—comparable, that is, to a composed score—then existing analytical models can be used. Steve Larson and Steven Block have

both taken this approach. In “Schenkerian Analysis of Modern Jazz,” Larson applies Schenkerian analysis to improvisations by Thelonious Monk, Bill Evans, and Oscar Peterson, uncovering large-scale linear continuities.⁴ This approach works especially well when considering improvisations by Bill Evans, who has acknowledged in interviews a tendency to conceive of his improvisations in terms of long-range voice-leading.⁵ Block’s “Pitch Class Transformation in Free Jazz,” on the other hand, uses the tools and techniques of set theory to uncover harmonic and melodic organization in improvisations of Ornette Coleman, John Coltrane, Cecil Taylor, and others.⁶ Figure 1.8 reproduces Block’s analysis of the opening of Coltrane’s improvised solo on the title track of the recording *Ascension*.⁷ Block does an excellent job of demonstrating how Coltrane’s improvised solo can be heard in terms of manipulations and transformations of pitch-class sets. Specifically, he argues that Coltrane generates the first half of this excerpt (mm. 1–16) by using two similar trichords (3-2 and 3-3), while in the second half (mm. 17–24), he switches to stringing together various transpositions of trichord 3-7.⁸ Furthermore, he says that the composite tetrachord created by combining the specific 3-2 and 3-3 trichords of the first half results in a tetrachord (4-3) that can be transformed into the tetrachord (4-26) created by combining the first two 3-7 trichords of the second half.

Although the arithmetical rigor of Block’s analysis is impressive, by focusing strictly on quantitative set-theoretical issues he tends to ignore more qualitative musical issues. Besides contrasting the pitch-class content of the first and second halves of the improvisation, one could also compare their difference in musical “character”: in the first half, Coltrane seems to be thinking rhythmically, repeatedly grouping eighth notes in threes, creating a cross-rhythm with the clearly stated 4/4 meter provided by the rhythm section. In the second half, when he switches to basing his improvisation on the 3-7 trichord, his improvisation also changes in character, featuring longer, arcing melodic gestures. A more holistic analysis might consider the relationship between the change in the quantitative materials (the pitch-class sets) and this qualitative change in character. Furthermore, considering the rhythm section’s accompaniment might lend additional insight into how Coltrane generates his improvisation. For example,

Section 1
transcription

analysis

etc.

14 16 3 24 3 3-7 3-7 3-7 3-7 *4-26

Figure 1.8. Steven Block's analysis from "Pitch Class Transformation in Free Jazz."

toward the end of the first half of the excerpt, pianist McCoy Tyner begins repeating the whole-step tone cluster $A\flat-B\flat$; it is Coltrane's subsequent emphasis on these same pitches that facilitates the change to trichord 3-7 beginning in m. 17.⁹

Other writers take a different approach, viewing improvisation as something intrinsically different from composed music, and focus their analyses on the improvisational process. In "A New Look at Improvisation," Ed Sarath describes the main difference between composition and improvisation as one of temporality:

[In composition,] the composer generates the material in one time frame and encodes them in a work in another. . . . The composer enters the 'timescape' of a work, and yet may also step back to isolate, reflect on, and possibly revise any given moment. . . . By 'improvisation,' I mean the spontaneous creation and performance of musical materials in a real-time format, where the reworking of ideas is not possible.¹⁰

He argues that because of the fundamental difference in temporal orientation of composition and improvisation, analytical models created for the examination of composed music are not as useful for the analysis of improvisation. Perhaps the author to deal most extensively with the product/process distinction is Charles Keil. In "Motion and Feeling through Music," first published in 1966, Keil argues for an analytical approach to jazz performance that focuses on musical *process*.¹¹ This article was written in response to Leonard Meyer's *Emotion and Meaning in Music*,¹² which, according to Keil, approaches music entirely in terms of syntax:

[Meyer] develops his thesis by first examining the form of music, a succession of tones, and then relating this form via psychological principles to meaning and expression. This procedure . . . implies not only a one-to-one relationship between syntactic form and expression but a weighting in favor of the former factor to the detriment of the latter. This tight equation of form and expression that for Meyer equals "embodied meaning" yields excellent results when applied to the generally through-composed and harmonically oriented styles of our own Western tradition. . . . When, however, this equation and the corresponding evaluative criteria are applied to non-Western styles . . . we often find that something is missing.¹³

	Embodied Meaning	Engendered Feeling
1. Mode of Construction	composed	improvised
2. Mode of Presentation	repeated performance	single performance
3. Mode of Understanding	syntactic	processual
4. Mode of Response	mental	motor
5. Guiding Principles	architectonic (retentive)	"vital drive" (cumulative)
6. Technical Emphases	harmony/melody/ embellishment (vertical)	groove/meter/ rhythm (horizontal)
7. Basic Unit	"sound term" (phrase)	gesture (phrasing)
8. Communication Analogues	linguistic	paralinguistic (kinesics, proxemics, etc.)
9. Gratifications	deferred	immediate
10. Relevant Criteria	coherence	spontaneity

Figure 1.9. Keil's Table of Contrasts.

What Keil finds missing in Meyer's syntactically focused notion of "embodied meaning" is any substantive consideration of musical process, and in response, Keil proposes an alternative set of musical characteristics that contribute to what he calls "engendered feeling." He compares several differences between these two aspects of musical experience in the table of contrasts reproduced in figure 1.9 (338).

Keil uses this list to contrast specific characteristics of European music and what he calls an "African or African-derived music" such as jazz. He says that European music, being composed, lends itself to repeated performances; jazz, on the other hand, being improvised, cannot be repeated exactly, and therefore can exist only as a single performance. In items 3 and 4 on his list, he seems to reify a mind/body dualism, saying that European music creates a mental response through syntactical organization, while jazz creates a more physical, motor response through musical process. He clarifies this in items 5 and 6, saying that "in composed music the structure or architecture is obviously of great importance; broadly speaking, melody rests upon harmony and embellishment upon melody" (344). Jazz, in contrast, places primary importance on the "vital drive," or "groove" that the musicians create (Keil's explanation of the ways that musicians create "vital drive" and "groove" will be examined more closely later in this chapter). In items 7 and 8, he again develops his mind/body dualism, saying that European music—with its emphasis on the syntactical content of a phrase—lends itself to linguistic analogy, while jazz—which places less emphasis on the *content* of a phrase and more emphasis

on the gesture and sense of movement it creates—is better described in terms of kinesthetics. Finally, in items 9 and 10, he contrasts the ultimate goals of both kinds of music, saying that in European music “resolutions must be anticipated and patiently awaited, gratifications must be deferred” (346), and that the ultimate arrival of these resolutions gives a composition a sense of coherence. On the other hand, he says that since jazz’s guiding principle of “vital drive” is present continuously throughout a performance, the gratification is not deferred, but immediate, that this lack of deferred gratification relaxes the need for any long-range sense of closure, and that, therefore, jazz performers place more emphasis on the perceived spontaneity of a given gesture rather than on its coherence to the performance as a whole.

Keil acknowledges that whereas most music involves aspects of both syntactical “embodied meaning” and processual “engendered feeling,” certain kinds of music lean one way or the other; specifically, he says that the music of the Western tradition tends toward embodying meaning, while the primary goal of an “African-derived” style such as jazz is to engender feeling. Therefore, he argues, an analysis that focuses exclusively on structural, syntactical aspects of a jazz performance overlooks the most important aspects of the performance: the way that it engenders feeling through subsyntactical musical processes. I feel that in criticizing Meyer’s focus on musical syntax, Keil perhaps swings too far in the other direction, narrowly focusing on—and overemphasizing almost to the point of fetishizing—subsyntactical processes, and as a result ignores the fact that, while subsyntactical aspects are an important part of a jazz performance, jazz musicians also use syntactical means such as goal-oriented harmonic progressions and melodies, motivic development, and recurring phrase structures to create an “architectonic” sense of deferred gratification and coherence.

Before defining my perspective on this issue, I’d like to briefly discuss Jean-Jacques Nattiez’s ideas regarding the nature of the musical object as presented in *Music and Discourse: Toward a Semiology of Music*.¹⁴ Nattiez says that a musical work is “three dimensional”—it consists not only of (1) a score, which he calls the “trace,” but also (2) the compositional processes that led to its creation, as well as (3) the perceptual/cognitive responses of the listener. Figure 1.10 reproduces Nattiez’s diagram of these three music-ontological dimensions. In this

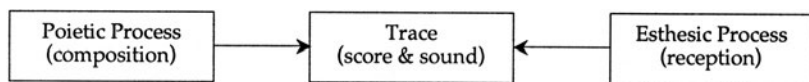


Figure 1.10. Nattiez's three dimensions of a musical work.

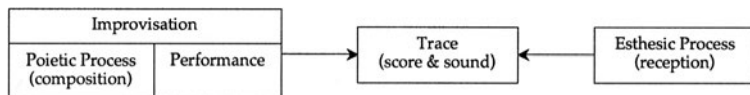


Figure 1.11. Nattiez's model modified to describe jazz.

diagram the Poietic (or compositional) process produces the “trace,” or score, which is realized by a performer or group of performers. The resulting performance is then heard by the listener, who perceives and cognizes the performance through Esthetic processes. Nattiez’s ideas provide a nice compromise in the argument over whether jazz improvisation should be viewed as a product or a process. According to Nattiez, music is *both* product *and* process; a musical analysis should not only examine the trace (or product), but also consider the processes that led to its creation as well as a listener’s perceptual and cognitive response.

Figure 1.11 modifies Nattiez’s diagram to more accurately describe the “three dimensions” of a *jazz* performance. Since jazz is substantially improvised in performance, the poietic process and the performance occur at the same time in the same person—in other words, the musician composes and performs simultaneously without the intermediate step of writing the improvisation down. As a result, the only trace is the sound produced by the performer, which is received by the listener, initiating the esthetic process of perception and cognition.

In order to adequately describe the perspective of the jazz musician, the diagram needs one further modification, shown in figure 1.12. In small-group jazz, all of the musicians are improvising simultaneously. The additional arrow in this example accounts for the fact that besides being a composer and performer each musician in the group is also a *listener*—that is, they are hearing the traces produced by the other musicians, and deciding whether or not they will alter their improvisations based on what they hear. Thus, this diagram describes a (not quite) closed loop that is occurring around each musician: the

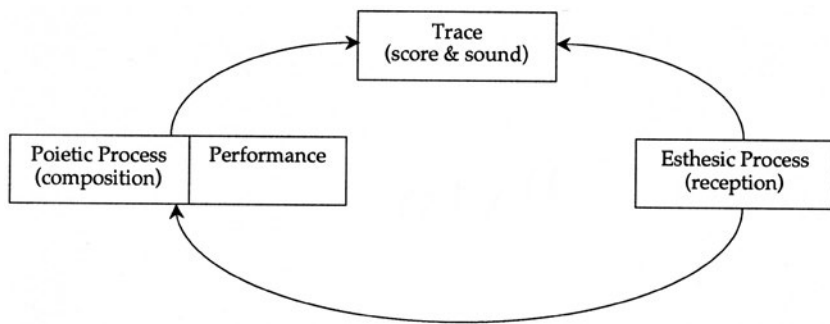


Figure 1.12. Nattiez's model modified to describe improviser's perspective.

jazz musician decides what to improvise, plays it, hears the other musicians improvising, at which point the process starts all over again, in almost immediate succession. In the next section, “Musical Roles and Behaviors,” I will consider specific constraints that each member of a small jazz ensemble has over the ways that they combine poietic and esthetic processes in the generation of their improvised parts.

To demonstrate the insights we can gain through adapting Nattiez's model to describe a jazz performance, I would like to briefly examine and critique a published analysis of Miles Davis's improvised solo on the tune “So What.” The performance is from the 1959 album *Kind of Blue*,¹⁵ and the analysis is from William Thomson's 1998 *College Music Symposium* article entitled “On Miles and the Modes.”¹⁶ Thomson analyzes Davis's solo in order to critique the concept of “modal jazz,” a term often used to describe jazz performances in which long sections of music are based on a “mode” rather than a series of chords that may move through several different key centers (modal jazz will be discussed more thoroughly in chapter 4). The tune “So What” is often considered paradigmatic of modal jazz; as shown in figure 1.13, it is a thirty-two-bar AABA form, in which the A sections are based on the D dorian mode, whereas the B section is based on E $\flat$ dorian.

Thomson says that while the concept of modal jazz may have influenced the creation of the recorded performance of “So What,”

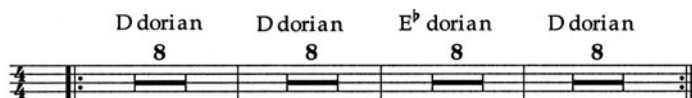


Figure 1.13. “So What,” formal sketch.



Figure 1.14. “So What,” Miles Davis’s solo, mm. 1–16.

whether a listener can perceive this performance as “modal” is another matter. He begins by examining Davis’s first improvised chorus, transcribed in figure 1.14. He says that in this chorus, Davis begins “with the best of dorian intentions” by emphasizing D and A in the first sixteen measures of his improvisation, and that by doing this he helps to project the “aura” of D dorian (21).

The crux of Thomson’s argument appears when he considers the first eight measures of Davis’s second improvised chorus, transcribed in figure 1.15. Here, he says, Davis’s strong emphasis on the pitches C, E, and G destroys the sense of D as a “prevailing nucleus,” and he describes this section as sounding more like “unrequited” C major than D dorian (25). He argues that not only does Davis’s emphasis on the C major triad weaken the overall sense of D-ness, but also this emphasis on C major is so strong that a listener *cannot* perceive the melody at this point as being in D dorian. He then goes on to make the broader argument that the concept of modal jazz, although perhaps useful as a compositional tool, doesn’t necessarily reflect the way a listener hears



Figure 1.15. “So What,” Miles Davis’s solo, mm. 32–41.

The image displays a musical score for the first eight measures of Miles Davis's solo in the jazz standard "So What." The score is written for four instruments: Trumpet, Piano, Bass, and Drums. The key signature is B-flat major (two flats) and the time signature is 4/4. The measures are numbered 1 through 8 above the staff lines. The Trumpet part features a melodic line with various intervals and rests. The Piano part provides harmonic support with chords and single notes. The Bass part plays a steady eighth-note pattern. The Drums provide a consistent rhythmic foundation with a pattern of eighth notes.

Figure 1.16. "So What," Miles Davis's solo, mm. 1–8.

the music, and therefore is of limited usefulness when analyzing or theorizing a listener's perception of this kind of performance.

To rephrase Thomson's argument in terms of Nattiez's model, there is a putative disparity between the performers' poietic and the listener's esthetic processes; that is, the poietic process of basing an improvisation on a specific mode results in a sonic trace that doesn't necessarily convey that process to the listener. A crucial point that Thomson's analysis and argument overlooks is that the transcription of Davis's individual improvised melody is not an objectively "accurate" representation of the sonic trace heard by the listener. Rather, the listener hears Davis's solo within the musical context provided by the rhythm section, which consists of pianist Bill Evans, bassist Paul Chambers, and drummer Jimmy Cobb. Examining a transcription of all four musicians' improvised parts can give a better idea of this sonic context. In figure 1.16, which transcribes the first eight measures of Davis's solo, we can see how each of their parts relates to the prevailing

The musical score for Miles Davis's solo in "So What" (measures 32-41) is presented in two systems. The first system covers measures 32 to 36, and the second system covers measures 37 to 41. The instrumentation includes Trumpet, Piano, Bass, and Drums. The key signature is one flat (B-flat), and the time signature is 4/4. The Trumpet part shows a melodic line with some chromaticism. The Piano part features complex voicings with 4ths and 3rds. The Bass part plays a steady walking line. The Drums provide a consistent rhythmic pattern.

Figure 1.17. "So What," Miles Davis's solo, mm. 32–41.

D dorian mode: Paul Chamber's bass line walks up and down the mode with occasional chromatic passing notes, while Bill Evans's piano voicings colorfully combine 4ths and 3rds, planing pandiatonically through the pitches of the D dorian mode. In this excerpt, all of the musicians are working within D dorian, and their improvised parts all help to convey that to the listener.

If we turn our attention to a full transcription of the beginning of Davis's second chorus, shown in figure 1.17, we see how Thomson's argument begins to falter. As Davis begins to wander away from D and A, basing his improvised melody on the pitches C, E, and G, the other members of the rhythm section counter this by ever more strongly emphasizing D: Paul Chambers plays a bass ostinato made up of pitches D, F, and A, and Bill Evans's piano voicings more explicitly define D dorian, being made up of a D minor triad with the addition of an alternation between B $\flat$ and C. Again, Thomson insists that

The image shows a musical score for Miles Davis's solo in the piece "So What". It consists of two systems of staves. The first system covers measures 28 to 32, and the second system covers measures 33 to 37. The top staff is for the Trumpet, and the bottom staff is for the Piano. In measure 32, a box highlights a specific melodic phrase in the trumpet part, and a line connects it to a similar phrase in the piano part in measure 36. The piano part features a steady bass line with chords, and the trumpet part features a melodic line with improvisation. The key signature is one flat (B-flat major/D minor), and the time signature is 4/4.

Figure 1.18. “So What,” Miles Davis’s solo, mm. 28–37.

because Davis’s improvised line centers on C, E, and G, it *cannot* be perceived as being a D dorian melody. On the contrary, I think that given the strong emphasis on D by the piano and bass, Davis’s line cannot be heard as anything *but* a melody in D dorian—a somewhat dissonant, colorful melody to be sure—but nevertheless a melody consisting of the 7th, 9th, and 11th of a tertian stacking of the pitches of the D dorian mode, rather than the root, 3rd, and 5th of C major.

Whereas we have been focusing on the performer’s poietic processes relative to a (nonperforming) listener’s esthetic processes, refocusing this analysis toward the perspective of the musicians taking part in the performance also yields interesting results. As we saw in the modification of Nattiez’s diagram to reflect the improvisational process in figure 1.12, improvising musicians engage in a dynamic, interactive process of deciding what to play, playing it, hearing what the other musicians are playing, and adapting what they play to take into account the other musicians’ improvisations. Such give-and-take between musicians seems to take place in this performance of “So What.” In figure 1.18, which transcribes the measures leading up to Davis’s second chorus—that is, the chorus in which he bases his improvisation on the pitches C, E, and G—we can see a series of musical events in which the musicians seem to respond to one another. Three

measures before the beginning of this second chorus, Bill Evans plays a chord voicing containing C, E, and G as its top three pitches. Since this is the first time in the performance that he plays this voicing, it's possible that Davis's subsequent emphasis on these three pitches is a response to Evans. In fact, beginning at this point, we can theorize an interesting chain of interactions: Evans plays a voicing emphasizing C, E, and G, which Davis registers, as it were, improvising a melody containing these pitches, which in turn affects the way Evans and Chambers improvise their accompaniment to this melody: they strongly emphasize D dorian in order to balance out this melody's tonal ambiguity.

In adapting Nattiez's model to describe the perspective of a jazz musician, I've laid the groundwork for my views on the process/product question and set up the analytical perspective that will be used throughout this work: I will analyze the product of an improvisation (in the form of a transcription) in order to deduce processes that contributed to its generation. The analyses will attempt to bridge the gap and relax the tension between Keil's "embodied meaning" and "engendered feeling"; that is, they will explore the ways that improvising jazz musicians balance the constraining factors of the structural, syntactical aspects of the composition they are performing with the dynamic, interactive processes that take place between the members of the ensemble.

Since the analyses will focus on improvisational process, the music will be examined as an "in-time" phenomenon—musical events are considered in the order in which they occur. The analyses will, in effect, slow down the performance process and consider the effects of musical events as they occur, especially focusing on the transfer of ideas between players in real time. This kind of approach is similar to that of certain strains of phenomenology,¹⁷ except that the focus of phenomenology tends to be the listener's perceptions, while the focus of my analyses will be the relationship between what the jazz musician hears and how this affects his or her improvisation—the two concerns are not perfectly concentric.

Much has been written about individual jazz musicians' improvisations; however, the musical effects of interaction between improvisers has not received much attention. Notable exceptions include the

writings of Paul Rinzler, Paul Berliner, and Ingrid Monson. Each of these authors describes and analyzes musical interaction between players in jazz performance. In "Preliminary Thoughts on Analyzing Musical Interaction among Jazz Performers," Rinzler lists five types of common interactions, including: (1) "call and response," which he describes as musical dialogue between equals; (2) "fills," which he says are similar to call and response, except that in this case a musician briefly emerges from the background of the performance to respond to the foreground soloist's improvisation; (3) "accenting the end of formal units," which he describes as a coordination between the soloist and rhythm section to mark structural boundaries; (4) "common motive," which he says involves the "exact repetition of a phrase"; and (5) "responding to the 'peaks' of the soloist," which he describes as the rhythm section's support of the overall level of intensity of a soloist's improvisation.¹⁸ Whereas this list does indeed describe some common ways that jazz musicians interact, Rinzler seems to view these interactions as isolated techniques that the musicians occasionally draw upon and incorporate into a performance. In contrast, I will be viewing interaction as something more elemental, as more of a continual musical process that is incorporated into the very fabric of the performance, and which may, at times, manifest itself in the form of "fills" or "call and response," but which may also involve much subtler kinds of interactions, involving, say, the relationship between a soloist's improvised melody and a pianist's chord voicings, or the effect of a drummer's rhythmic patterns on a soloist. In other words, by viewing a jazz performance in terms of the model of the improvisational process that I have outlined above, anything played by any member of the ensemble can potentially have an effect on any other member of the ensemble, increasing the range of possibilities for potential interactions far beyond those described in Rinzler's list.

Berliner and Monson take a more ethnomusicological approach in their works. Berliner's *Thinking in Jazz* is a comprehensive and appropriately ethnographic work, focusing on the role of the jazz community in a jazz musician's acquisition of skills as a soloist and ensemble player.¹⁹ In *Saying Something*, Monson's goal is to show the relationships and similarities between musical interaction and cultural or social interaction.²⁰ She interviews musicians, analyzes the metaphors they

use to describe what's happening when they perform, and maps these descriptions onto transcriptions of jazz performances. Her stated goal is to explore the concept of interaction on several levels, including: "(1) the creation of music through the improvisational interaction of sounds; (2) the interactive shaping of social networks and communities that accompany musical participation; and (3) the development of culturally variable meanings and ideologies that inform the interpretation of jazz in American society" (2). Monson begins with an examination of mediated social and cultural practices, and then looks for analogous principles in jazz performances. This book complements Monson's social and cultural emphasis by focusing on performance as an immanent experience. It will examine and explore the interactive musical processes between improvising musicians that contribute to the generation of a jazz performance.

The analysis of Cannonball Adderley's improvisation at the beginning of this chapter briefly touches upon these issues. First, the improvisation was considered in isolation and described in detail. Questions were then raised as to the reason for a drastic change in the style of the improvisation. The improvised line itself did not really supply any answers to these questions, but an examination of Wynton Kelly's simultaneous improvisation suggested a strong possibility: Adderley's change of style was in response to Kelly's fleeting reference to that style. A further example demonstrates more interaction between the players, but in the opposite direction—this time, a motive played by Adderley influences Kelly's improvised accompaniment. Thus, the analysis moved from a consideration of the *product* (the improvisation and its transcription) to a deduction of the interactive *processes* that influenced its creation.

These issues are examined in more detail over the course of this work. In the remainder of this chapter, each member of the rhythm section will be considered, as will common relationships and interactions between various combinations of rhythm section members and the soloist. We will see that, even though the interaction between musicians can have a strong influence over how each musician realizes his or her part, musicians are not *entirely* free to improvise whatever they want; rather, they each fulfill certain prescribed roles within the ensemble and for the most part improvise within the constraints of

these roles. Chapter 2, “Harmony and Interaction,” will explore the process of harmonic negotiation that takes place in a small-group jazz performance, examining ways that the interaction between musicians can alter or modify the realization of a tune’s harmonic progression. Chapter 3, “Form and Interaction,” will examine ways that musicians play within predefined phrase structures and interact to negotiate the definition of formal boundaries. Finally, Chapter 4, “Breaking Down the Boundaries: Steps toward Free Jazz,” will explore performances that connect bebop and free jazz by breaking down the constraining factors of prescribed ensemble roles, predefined harmonic progressions, and predetermined formal structures that were examined in Chapters 1–3. We will see that breaking down these constraints usually requires an increase and intensification in the interaction between players in a small jazz ensemble.

Musical Roles and Behaviors

Because the main concerns of this book are the ways in which jazz musicians communicate and interact in performance as well as the ways in which these interactions affect each musician’s improvisation, it’s necessary to define the normative performance roles and behaviors of each musician in a small jazz group. Defining these roles will provide a basis for comparison; in order to discover and point out heightened levels of interaction, one must first understand the usual ways that standard jazz performance practice controls and allows for these interactions. It’s important to note that the following descriptions of the roles and behaviors in jazz performance are not rules or laws, but are the result of an informal agreement within the jazz community. This informal agreement allows for musicians who have never played together to perform without rehearsal—each musician knows and understands his or her normative role, and knows that if each participant fulfills their role, the performance will cohere.²¹

Figure 1.19 contains a transcription of an excerpt from Charlie Parker’s recording of his composition “Now’s the Time.”²² This recording provides a good example of standard jazz performance practice, and the description that follows will be used to define the normative

The musical score is presented in two systems. The first system includes staves for Trumpet, Sax, Piano, Bass, and Drums, covering measures 1 to 4. The second system continues with measures 5 to 8. The key signature has two flats (B-flat and E-flat), and the time signature is 4/4. The piano part is highly active, featuring dense clusters of notes and complex rhythms. The bass part maintains a steady eighth-note pulse. The drums provide a consistent rhythmic foundation with a mix of eighth and sixteenth notes.

Figure 1.19. “Now’s the Time.”

roles of each musician in a small jazz ensemble. This recording features a quintet composed of Charlie Parker (alto saxophone), Miles Davis (trumpet), Dizzy Gillespie (piano),²³ Curly Russell (bass), and Max Roach (drums). To respect the copyright provisions for “Now’s the Time,” the composed melody is not fully notated; after the opening motive (which begins with the pick-up to m. 9), only the rhythm of the melody is notated. The remainder of the performance in this excerpt (the piano, bass, and drum parts and the improvised saxophone melody) is included.²⁴

In standard jazz performance practice, a small group usually divides into two subgroups, the *front line* and the *rhythm section*. These

The musical score for Figure 1.19 (continued) spans measures 9 to 16. It is arranged for a jazz ensemble with five staves. The top two staves represent the front line (saxophone and trumpet), and the bottom three staves represent the rhythm section (piano, bass, and drums). The key signature is one flat (B-flat major or D minor). The time signature is 4/4. Measures 9-12 are marked with F7 and B^b7 chords, while measures 13-16 are marked with B^b7 and F7 chords. The front line plays a melodic line, while the rhythm section provides harmonic and rhythmic support.

Figure 1.19. (continued)

two subgroups divide up the musical responsibilities: the front line is usually responsible for the melody, whereas the rhythm section provides and defines the harmony and rhythm. These divisions are, of course, not absolute; the front line's melodies often contribute to the overall harmony and rhythm of the performance, and the rhythm section's harmony and rhythm often have melodic aspects or effects. Nevertheless, the division, however crude, serves a useful purpose: it symbolizes the group aesthetic of jazz, an aesthetic that values both the division of labor and the sharing of responsibilities. In this particular performance, the front line consists of saxophone

The musical score for Figure 1.19 (continued) spans measures 17 to 24. It is written in 4/4 time with a key signature of one flat (B-flat). The score is organized into two systems of four staves each. The first system (measures 17-20) features a melodic line in the top staff and a piano accompaniment in the bottom staff. The second system (measures 21-24) continues the melodic line and piano accompaniment. The drum part is indicated by 'x' marks on a single staff at the bottom of each system. The chord changes are indicated above the staves: G-7 (measure 17), C7 (measure 18), F7 (measure 19), C7 (measure 20), F7 (measure 21), Bb7 (measure 22), F7 (measure 23), and an unmarked measure (measure 24).

Figure 1.19. (continued)

and trumpet, and the rhythm section is made up of piano, bass, and drums. The specific instrumentation of a small jazz ensemble may vary—the front line might consist of only a saxophone or a trumpet, it might include a trombone or additional saxophonists, the rhythm section might have a guitarist in addition to or instead of a pianist—but the group of instruments used on this recording is one of the most common in small-group jazz.

The large-scale form of this performance is determined by a standard performance practice known as the *head arrangement*. A head arrangement begins with a statement of a composed *tune* (the “head”), follows with a series of improvised solos by some or all of the members of the group, and ends with a restatement of the head.²⁵ In this performance,

25 26 27 28

B^b7 F⁷ D⁷ 3

29 30 31 32

G⁷ C⁷ F⁷ 3

Figure 1.19. (continued)

the head arrangement is preceded by an eight-measure introduction played by the rhythm section. After this introduction, the saxophone and trumpet enter in m. 9 with a statement of the head, in this case, Charlie Parker's twelve-bar blues composition "Now's the Time."²⁶ Being twelve measures long, this statement of the head extends through m. 20. The head is then followed by improvised saxophone and trumpet solos, and finally ends with a restatement of the head. The behavior of the saxophone and trumpet in this performance is standard for the front line: being primarily responsible for the melody, they play the head, improvise new melodies, and finally repeat the head to end the performance.

The primary role of the rhythm section is to accompany, support, and complement the melody of the front line by providing and

defining harmony and rhythm. Reflecting the clear division of labor *between* the front line and the rhythm section, there is also a division of labor *within* the rhythm section. The three members of the rhythm section—piano, bass, and drums—cooperate and collaborate in interesting ways to define the harmony and rhythm. They do this by fulfilling three different functions: (1) the defining of the harmony, (2) the defining of regular rhythm (pulse and meter), and (3) the use of nonregular rhythmic events to support the melody or propel the music forward. No single member of the rhythm section is responsible for any one of these three functions. Instead, each player normally contributes to two of the three functions. In order to define the roles of each member of the rhythm section and the ways that they contribute to the definition of harmony and rhythm (both regular and nonregular), I'd like to examine each instrument in turn, basing my examination on the transcription in figure 1.19.

The Bassist

The role of the bass in the rhythm section is both rhythmic and harmonic. Beginning in m. 9 of figure 1.19, the bass player provides the basic quarter-note pulse of the performance by means of a *walking bass line*, which, once begun, continues through to the end of the recording. Besides defining the basic pulse, the bassist also contributes to the realization of the harmonic progression by improvising a line that emphasizes harmonically significant pitches (usually roots, 3rds, or 5ths) on beat 1 or 3 of each measure. For example, in m. 9, the bass player plays the root of F^7 (F) on beats 1 and 2 before moving to the 3rd (A) on beats 3 and 4. Whereas the rhythmic and harmonic aspects of the bass line are the most important, its melodic aspects should not be overlooked. The bass player's improvised line not only emphasizes harmonically significant pitches on strong beats, but also connects these pitches through the use of passing and neighbor notes. In m. 10, the bass line begins on the root of Bb^7 , then plays a $B^{\natural}$ passing note that connects to C (the 5th of F^7) on the downbeat of m. 11. This process—improvising a quarter-note line that melodically connects harmonically significant pitches—continues throughout. Even though

this process seems somewhat restrictive, the actual choice of which notes to play is left up to the bassist, and this choice can be influenced by the other band members' improvisations. For example, the bass player's decision to play the 7th of F^7 ($E\flat$) on the downbeat of m. 24 may have been influenced by the fact that Charlie Parker ended his phrase on an $E\flat$ in the previous measure. It is important to note that a bassist isn't always restricted to playing a walking bass line, but this is simply its most common, normative role in small-group jazz.

The Drummer

The drummer helps the bass player define the basic, regular rhythm by playing two different repetitive patterns: the *ride pattern* and the *backbeat*. These two patterns are notated in figure 1.20a. The ride pattern is played on the suspended cymbal—which, for this reason, is often called the “ride” cymbal. Although the ride pattern is notated as a quarter note followed by two eighths, in practice the two eighth notes are not played evenly, but are “swung”; that is, they are played unevenly—the first note longer, the second note shorter. A more rhythmically precise notation of the ride pattern might be something like that contained in figure 1.20b, although the exact relationship between the long and short notes may vary depending on a variety of factors, including tempo and style.²⁷ Besides the ride pattern, the drummer also plays another regular rhythmic pattern called the *backbeat*. This

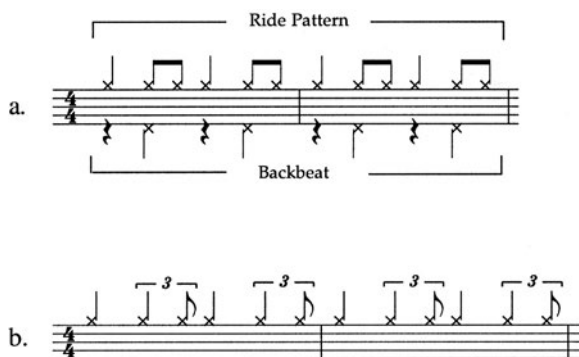


Figure 1.20. Basic drum patterns.

pattern is played on the “hi-hat,” which consists of two small cymbals on a stand that are attached to a foot pedal. When the drummer presses down on the foot pedal, the two small cymbals come together, making a short, crisp, dry sound often rendered onomatopoeically by the word “chick.” To play the backbeat, the drummer depresses the foot pedal on beats 2 and 4 of each measure.

In playing these two patterns, the drummer contributes to the definition of the regular rhythm of the performance: the ride pattern defines the quarter-note pulse; the backbeat helps to define the meter.²⁸ Although both of these patterns are played on cymbals, they sound very different; because the drummer typically allows the ride cymbal to ring, only the attack of the ride pattern is heard, and the ringing of the cymbal provides a sustained sonic backdrop for the performance. This contrasts with the short, percussive sound of the hi-hat backbeat that overlays the ride pattern. The result is that the ride pattern creates a sense of continuity, flow, and forward motion, whereas the backbeat segments that flow into metrically emphasized units and acts as an undertow, generating tension within the forward flow. Returning to figure 1.19, the drummer establishes both the backbeat and ride pattern immediately in the introduction, and continues them throughout the entire performance.

Besides contributing to the regular rhythmic pulse and meter, the drummer also supports the melody and improvisations of the front line by playing nonregular rhythms on the snare drum, tom-toms, bass drum, or crash cymbal. Typically, the drummer does this in one of three ways: (1) by *reacting* to the front line, (2) by *coordinating* or *locking up* with the front line, or (3) by *anticipating* or *setting up* the front line’s melody or improvisation. Figure 1.21 contains the drum part and the front line melody of mm. 9–20. In this figure, the drummer provides all three types of rhythmic support. In mm. 9, 10, and 12, the drummer *reacts* to the front line by hitting the snare drum immediately after the melody reaches each quarter-note resting point. These accented, syncopated rhythms provide a rhythmic vitality that propels the music forward. In the first two measures of the second four-bar phrase (mm. 13 and 14), the drummer plays the same rhythm as the previous phrase, but this time, because the melody is one note longer, the

The musical score is presented in three systems, each with four measures. The first system (measures 9-12) shows the Trumpet and Saxophone parts with a steady eighth-note accompaniment. The drums play a consistent pattern of eighth notes. The second system (measures 13-16) continues the melody, with the drums providing a steady beat. The third system (measures 17-20) shows the melody changing, with the drums continuing their pattern. Arrows point from the drum hits to the syncopated entrances of the front line melody in measures 10, 14, 18, and 20.

Figure 1.21. “Now’s the Time,” front line and drums, mm. 9–20.

drummer and the front line *lock up*. Even though the overall musical effect is similar to the first phrase—that is, the music is propelled forward—the relationship between the front line and the drummer is one of coordination and cooperation, rather than reaction. In mm. 16 and 18, the drummer *sets up* the syncopated entrance of the melody by hitting the snare drum squarely on the beat. In these measures, it’s the front line’s syncopated entrance that provides the rhythmic vitality, but the drummer contributes to this vitality by providing a frame of reference (the squarely on-the-beat quarter-note snare-drum hit) that sets off and defines the syncopated entrance of the front line.

The Pianist

The role of the pianist in the rhythm section, like that of the bassist, is both rhythmic and harmonic. The pianist contributes to the definition of harmony and rhythm by comping²⁹—that is, by playing chord voicings that define the harmonic progression of the tune in a rhythmic, often syncopated manner. After the eight-measure introduction, the pianist comps throughout, for instance, as in figure 1.19. The rhythmic placement of the pianist's chords is nonregular, and these chords interact with the front line in ways analogous to that of the drummer's snare drum. In other words, the pianist reacts to, locks up with, or sets up the melodies of the front line.

Musical Relationships within the Rhythm Section

As discussed earlier, the three members of the rhythm section support the front line by defining the harmony, the regular rhythm, and non-regular rhythmic events. The preceding descriptions of the normative role of each member of the rhythm section do not give the complete picture, however. To gain a more complete understanding of how a rhythm section works, it's necessary to explore the network of relationships between the players. Figure 1.22 presents a schematic of these relationships. The lines connecting each pair of instruments show that each pair works together to define one of the three parameters. I'd like to briefly consider each pair of instruments and examine how they collaborate.

Bass and Piano

Figure 1.23 contains the bass and piano parts from mm. 9–12 of “Now's the Time.” As diagrammed in figure 1.22, these two instruments work

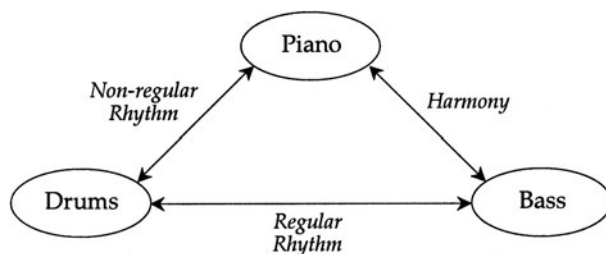


Figure 1.22. Schematic of relationships within rhythm section.

This musical score shows the piano and bass parts for measures 9 through 12 of the jazz standard "Now's the Time." The key signature is B-flat major (two flats) and the time signature is 4/4. The piano part is written on a grand staff (treble and bass clefs). In measure 9, the piano plays a chord voicing for F7 (root, 3rd, 5th, 7th, 13th). In measure 10, it plays a voicing for Bb7 (3rd, 5th, 7th, 9th). In measure 11, it plays a voicing for F7. The bass part is written on a single bass clef staff. In measure 9, the bassist plays the root and 3rd of F7. In measure 10, the bassist plays the root and 3rd of Bb7. In measure 11, the bassist plays the root and 3rd of F7. In measure 12, the bassist plays the root and 3rd of F7. Arrows point from the piano's chord voicings to the corresponding bass lines, indicating the harmonic relationship.

Figure 1.23. "Now's the Time," piano and bass, mm. 9–12.

together to define the harmony of the tune. In m. 9, the bassist plays the root and 3rd of the chord F^7 , and the pianist plays a voicing that contains the root, 3rd, 5th, 7th, and 13th. In m. 10, the pianist's voicing of B^b7 contains the 3rd, 5th, 7th, and 9th, but does not contain the root. The bassist, however, does provide it, so that between the pianist and bassist the complete B^b7 is spelled out. Because there is a certain amount of flexibility in the realization of the harmonic progression, the bassist and pianist must always listen to each other in order to effectively and musically define the harmony. Chapter 2 explores the ways that interaction between the piano, bass, and front line affects the realization of the harmonic progression.

Bass and Drums

Figure 1.24 contains the bass and drum parts from mm. 9–12 of "Now's the Time." As seen in the diagram of figure 1.22, these two instruments work together to define the regular rhythm (i.e., the pulse and meter) of the performance. The walking bass line and the ride pattern both provide the basic quarter-note pulse, and the hi-hat's backbeat helps to define the meter. In order for the performance to

This musical score shows the bass and drum parts for measures 9 through 12 of "Now's the Time." The key signature is B-flat major (two flats) and the time signature is 4/4. The bass part is written on a single bass clef staff. In measure 9, the bassist plays a walking bass line (root, 3rd, 5th, 7th). In measure 10, the bassist plays a walking bass line (root, 3rd, 5th, 7th). In measure 11, the bassist plays a walking bass line (root, 3rd, 5th, 7th). In measure 12, the bassist plays a walking bass line (root, 3rd, 5th, 7th). The drum part is written on a single staff. The ride cymbal plays a steady quarter-note pulse. The hi-hat plays a backbeat pattern (quarter note, eighth note, eighth note, quarter note). The bass drum plays a pattern of quarter notes and eighth notes. Arrows point from the bass line to the corresponding drum patterns, indicating the rhythmic relationship.

Figure 1.24. "Now's the Time," bass and drums, mm. 9–12.

“swing,” the drummer and bassist must listen intently to each other and coordinate their rhythmic patterns.

In the abstract, these rhythmic patterns played by the bassist and drummer are very simple, but when one considers the *way* that the musicians play these rhythms and how this contributes to the feeling of swing, things get more complex. Returning to “Motion and Feeling through Music,” Charles Keil examines this issue in some depth, exploring the subsyntactical aspects of the relationship between bassists and drummers (341–344). He argues that the swing, or “vital drive”³⁰ of a jazz performance is created when the bassist and drummer pull against an underlying pulse, playing their rhythms either slightly ahead of or slightly behind the beat. Keil creates a taxonomy of both bassists and drummers, based on their relationship to the underlying pulse. He describes drummers as falling into two distinct groups: those who play “on top” of the beat, and those who “lay back” behind it (341). He makes a similar distinction between two basic styles of bass playing, defining one as “chunky”—that is, heavy, percussive, drumlike, on or ahead of the beat—and the other as “stringy,” which he describes as light, sustained, slightly behind the beat (343).

Keil considers a number of different combinations of bassists and drummers from various jazz ensembles. He describes Thelonious Monk’s 1958 rhythm section as consisting of a “chunky” bassist (Ahmed Abdul Malik) and an “on-top” drummer (Roy Haynes); he likewise describes Miles Davis’s 1956 and 1965 rhythm sections (bassists Paul Chambers and Ron Carter, drummers Philly Joe Jones and Tony Williams, respectively) as combinations of a “stringy” bassist and a “lay-back” drummer. Although this examination of the extremely subtle aspects of how the rhythm section musicians play their parts is interesting, insightful, and supports Keil’s argument that jazz is more about process than syntax, I feel that it yields mixed results. While Keil criticizes scholars such as Meyer who ignore the subtle, subsyntactical rhythmic aspects of music, he himself overgeneralizes and simplifies the subject by categorizing individual players into one of two groups, ignoring the fact that, in performance, bassists and drummers can change the way they play, modifying the placement of their rhythms in relationship to the beat in response to the musical circumstances. For example, drummer Tony Williams, whom Keil

describes as a “lay-back” drummer, actually has an extremely flexible and dynamic approach to rhythm, with the ability to place his rhythmic patterns ahead of, behind, or squarely on the beat.³¹ Likewise, bassist Paul Chambers, whom Keil describes as a “stringy” bassist, also can change the relationship of his bass line to an underlying pulse. In the Miles Davis Quintet’s performance of “Blues by Five,”³² for example, Chambers begins Davis’s solo in a relaxed manner, playing “in the pocket”—that is, just slightly behind the beat. As the performance progresses, John Coltrane begins to improvise, and as he builds the level of intensity throughout his solo, Chambers responds by closing the gap between his bass line and the underlying beat, moving more toward playing in the “chunky” style. In doing so, Chambers contributes to the increase in energy and intensity that Coltrane creates in his improvised solo.

Piano and Drums

Figure 1.25 returns to the performance of “Now’s the Time,” showing the piano and drum parts for mm. 9–12. In mm. 9, 10, and 12, the pianist and drummer lock up—that is, they both respond the same way to the front line melody. While the pianist and drummer play exactly the same rhythm in this example, this will not always be the case. Rather, the pianist and drummer share a musical function: to provide nonregular rhythmic events in response to, or in support of, the front line. Throughout the remainder of the example in figure 1.19, the pianist and drummer sometimes play the same rhythms, and sometimes do not. On the occasions that the pianist and drummer do lock up their rhythms, the effect is one of increased rhythmic vitality and drive.

The figure shows a musical score for piano and drums in 4/4 time, measures 9–12 of “Now’s the Time.” The piano part is written in treble and bass clefs. The drums part is written in a single staff with “x” marks for notes. Chords F⁷, B^b7, and F⁷ are indicated above measures 9, 10, and 11 respectively. Arrows point from piano notes to drum notes in measures 9, 10, and 12, indicating a “lock up” where both parts play the same rhythm.

Figure 1.25 “Now’s the Time,” piano and drums, mm. 9–12.

In “Now’s the Time,” the pianist and drummer lock up their rhythms in response to the melody; but it is also possible for this kind of interaction to go the other way, that is, the pianist and drummer might create an increase in musical intensity by coordinating their rhythms, which in turn may affect the way that a soloist improvises over their accompaniment. Just such an interaction takes place during trumpeter Nat Adderley’s solo on the Cannonball Adderley Quintet’s performance of “Spontaneous Combustion.”³³ Throughout the first part of his solo, Nat Adderley usually improvises simple, “bluesy” melodies composed of short, repetitive, rifflike figures. The musical effect created by this style of improvisation is one of relative stasis; the short, repetitive figures do not create a strong feeling of forward motion toward a musical goal. Approximately halfway through Nat Adderley’s solo, pianist Bobby Timmons sets up a new chorus by playing the strong rhythmic pattern transcribed in figure 1.26a. Timmons uses this pattern to signal a syncopated rhythmic pattern with which he continues his accompaniment at the beginning of the next chorus. With this syncopated pattern, which is transcribed in figure 1.26b, Timmons creates a tremendous feeling of forward motion by continually anticipating the strong beats (beats 1 and 3) by one half of a beat, placing his accented voicings on the second half of beats 2 and 4. As shown in figure 1.26c, drummer Louis Hayes, having heard Timmons’s signal in the previous two measures, responds immediately, locking up with Timmons’s rhythms.³⁴

This coordination between Timmons and Hayes further heightens the rhythmic intensity and strong sense of forward motion, and soloist Nat Adderley responds by changing the way he improvises. Figure 1.27 combines the transcription of Timmons’s and Hayes’s rhythmic patterns with Adderley’s improvised response. In contrast to the short, repetitive, non-goal-oriented melodies he improvised earlier in the solo, Adderley responds to the rhythm section’s strong feeling of forward motion by improvising a melody that itself flows strongly forward. His fast, ascending sixteenth-note scale rushes to its peak, and Adderley follows this opening gesture with a long descending line that flows all the way to the downbeat of m. 5. At this point, the tension and energy created by the rhythm section are resolved, and everything returns to normal: pianist Timmons returns to playing

Piano

Figure 1.26a. “Spontaneous Combustion,” piano comping pattern.

Figure 1.26b. “Spontaneous Combustion,” piano comping continued into new chorus.

Figure 1.26c. “Spontaneous Combustion,” lock-up between piano and drums.

The musical score is divided into two systems. The first system covers measures 1 through 5, and the second system covers measures 6 through 9. Each system includes staves for Trumpet, Piano, and Drums.

System 1 (Measures 1-5):

- Trumpet:** Measures 1-2 contain rests. In measure 3, the trumpet enters with a melodic line starting on a B-flat. The line continues through measures 4 and 5 with various eighth and sixteenth notes.
- Piano:** Provides a harmonic accompaniment using chords and single notes, primarily in the lower register.
- Drums:** Features a steady rhythmic pattern with eighth notes and accents.

System 2 (Measures 6-9):

- Trumpet:** Continues the melodic line from the previous system, showing improvisation with grace notes and slurs.
- Piano:** Continues the harmonic support with chords and moving lines.
- Drums:** Maintains the rhythmic foundation with consistent eighth-note patterns.

Figure 1.27. “Spontaneous Combustion,” trumpet solo with piano and drum rhythms.

sparse, syncopated voicings, drummer Hayes reverts to occasional rhythmic interjections, and Adderley returns to improvising simpler, less goal-oriented melodies.

Summary

This chapter describes small-group jazz performance as a complex interactive process. The basic, normative roles of each member of the ensemble are defined, as are common interactions between players. A general model of jazz improvisation, based on ideas of Jean-Jacques Nattiez, is proposed; this model defines the jazz musician as being simultaneously a listener, composer, and performer. That is, each musician improvises (improvisation being a combination of composition and performance), hears the other musicians' simultaneous improvisations, and adapts or modifies his or her improvisations based on what he or she hears. A more specific example of this model may help to clarify this process. In the case of the pianist, his or her basic role in small-group jazz is to comp—that is, to accompany the front line by playing the harmony of the tune in a rhythmic manner. There are a number of factors that influence the way that the pianist chooses to fulfill this role. The pianist must listen to the bassist and the drummer to hear the basic pulse and meter, must listen to the front line and the drummer to decide when to play the chords, and must listen to the bassist and the front line to decide how to voice the chords. In other words, what the pianist hears from the other musicians can (and should) have a very strong effect on what he or she plays. The opposite is also true, namely, what the pianist plays can have a very strong effect on how each of the other musicians fulfill their roles. The analysis of Cannonball Adderley's performance of "Groovin' High" demonstrated both of these possibilities: first, Wynton Kelly's comping affected Adderley's improvisation, then Adderley's improvisation affected Kelly's comping. This work will continue to explore the interactions between the players in small-group jazz by examining several transcribed jazz performances and will examine the ways that these interactions between players can affect each individual musician's improvisation.

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